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THE MANHATTAN PROJECT

CO- UNDER SECRETARY GENERAL:

KUZEY KARLIK

ACADEMIC ASSISTANT: EFE KOCOGLU

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1. Letter from the Secretary General

Dear Participants,

First of all, it is my utmost pleasure to serve you as the Secretary General for the first edition of ION Model United Nations Conference. I am more than proud for making this conference happen with my colleagues and looking forward to see every attendee.

Essentially MUN's are great opportunities to increase your consultation capabilities, your assertiveness and understand how policy is implemented. I can honestly state that MUN impacted my perspective on life in a positive way. For this reason, it is my mission to transfer the experiences I have gained to the new generation and to ensure that they affect them in the same way.

Lastly as the executive team, in order to put this conference together we really plod away. We really worked hard to burn the soul of IONMUN26. And do not forget ION is the timeless voices for peace.

Sincerely,

Doruk Sapmaz

2. Letter from Under Secretary General

Dear Delegates,

I would like to welcome you all to the Manhattan Project! I am Kuzey Karlık and it is my utmost pleasure to serve you as the under secretary general of this committee.

First, I would like to thank the executive team for granting me this opportunity in this marvelous conference. And I would like to thank my fellow Academic Assistant Efe Koođlu for his magnificent efforts and work ethic.

The study guide we wrote contains crucial information for this committee. Yet as its name states it's only a guide for you, so I am highly encouraging you to do your research properly, there will be a lot of debate points, disagreements and controversies hence you have to be ready.

I am sure this committee will be a blast, and I am looking forward to meeting you all on 3-5 August. If you have any questions you can always contact me through mail.

Sincerely,

Kuzey Karlık, Under Secretary General of the Manhattan Project,

kuzeykarlik@gmail.com

3. Letter from Academic Assistant

Dear delegates,

I am Efe Koçoğlu. I am a senior student at Muhittin Mustafa Böcek Anatolian High School. I had the delight of participating in Model United Nations conferences. These conferences taught me how to see from different perspectives, how to debate, and how to make instant decisions. I also gained valuable friendships in my MUN career.

I welcome you all to this glorious committee, the Manhattan Project. This was my dream committee, and my dear academic team, also my friend Kuzey, helped me through this journey. I really hope you have so much fun and learn new things from this committee. As the academic assistant of this committee, I want you to read this study guide very carefully. Don't hesitate to ask me; I am online 24/7. We expect wonderful things from you. There is no light without a shadow; there is no future without the past.

Efe Koçoğlu

Academic Assistant of The Manhattan Project

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4. Introduction to the Committee

The Manhattan Project was a top secret program to make the first atomic bombs during World War II. Its results had profound impacts on history: the subsequent nuclear arms race has radically changed the political world order in ways that are still evident today. Thousands of scientists, including theoretical physicist J. Robert Oppenheimer, took part in the Manhattan Project, often while they and their families were lodged at secret military bases in remote locations. It resulted in the two atomic bombs dropped on the Japanese cities of Hiroshima and Nagasaki in August 1945, which brought World War II to its end and probably killed more than 100,000 people. The Manhattan Project harnessed the enormous energy in the nucleus of the atom for the first time,” explains Cynthia Kelly, founder and president of the Atomic Heritage Foundation, a nonprofit dedicated to the history of the project and the atomic age. One of the project’s consequences was the creation of terrifying opposing arsenals of nuclear weapons. But it also resulted in innovations from medicine to space exploration and in the science and engineering of civilian nuclear energy, Kelly says. The U.S. Army Corps of Engineers created the Manhattan Engineer District in June 1942 to hide the development of the atomic bomb during the war hence that effort’s name of the “Manhattan Project.”

But historian of science Alex Wellerstein explains in an online overview that the project originated in an idea from the late 1930s that Nazi Germany might build an atomic bomb, so the U.S. should do so first. Historical records reveal that Germany didn't get far, but the prospect of a Nazi bomb was horrifyingly real. Several leading researchers worked for wartime Germany, including Werner Heisenberg. Even so, many scientists who favored an American atom bomb, including Hungarian-born physicist Leo Szilard, had fled the Nazis in Europe. In July 1939 Szilard and others enlisted the help of the renowned physicist Albert Einstein, then on holiday on Long Island, N.Y., to support them by writing a letter to President Franklin D. Roosevelt. The Einstein-Szilard letter, as it's known, changed history. It prompted Roosevelt to convene a committee to investigate the possibility of building an atomic bomb, and in 1941 this group became a new committee to lay the groundwork for the full project.

These early stages involved key contributions from the U.K. and Canada. But in the end, the atomic bomb was mostly an American weapon. After 1942 the Manhattan Project was the recognized Allied effort to build an atomic bomb. It mainly used uranium ore from a mine in what is now the Democratic Republic of the Congo, which was kept secret from the Germans. Otherwise, the project was conducted in the U.S., primarily at three top-secret towns: Oak Ridge, Tenn., where uranium was enriched until it was radioactive enough for nuclear fission; Hanford, Wash., where reactors transformed uranium into plutonium, an even more powerful nuclear fuel; and Los Alamos, N.M., where Oppenheimer directed the laboratory that designed and built experimental atomic bombs. There were also dozens of smaller sites. And officials went to extraordinary lengths to keep it all secret. World War II historian Alexandra Levy says most of the more than 600,000 people involved including the thousands of scientists, engineers, and technicians who worked on the weapons, as well as construction workers and the people who kept the three secret towns going were deliberately not told their purpose.

“Most of those people did not know that the goal of the project was to build a new type of bomb,” Levy says. “Today, between the Internet and social media, it’s difficult to imagine such a large scale endeavor remaining secret for long.”

Kelly adds, “We live in a very, very different world. Aside from one or two key senators agreeing to a blank check for the Manhattan Project, Congress and the press were kept in the dark. That would be impossible today.”

The Manhattan Project culminated in the Trinity test in New Mexico on July 16, 1945 the first detonation of a nuclear weapon. By that time, the U.S. had spent around \$2.2 billion the equivalent of around \$37 billion today.

But the dangers of a Nazi bomb had faded, and Japan was now the designated target. Although Japan never had an atomic bomb program, the idea of stopping its aggression with a show of awful destruction became fixed among Manhattan Project leaders, science historian Wellerstein says. He notes that Oppenheimer, then the charismatic director of the Los Alamos laboratory, twice voted in favor of the initial atomic bomb attack on Hiroshima which killed tens of thousands of civilians instead of a purely military target.

Oppenheimer is seen as essential to the success of the American atomic bomb project. “He contributed to some of the early scientific breakthroughs of the project,” Levy says. “His great gift was bringing together scientists, engineers and other technicians to collaborate on and solve problems.”

But Oppenheimer was also ambivalent about its results. In recalling his experience at the Trinity test in 1965, he quoted a story from the Hindu scripture the Bhagavad Gita about a prince, reluctant to kill his enemies, who witnessed the transformation of Krishna, an incarnation of the Hindu deity Vishnu: “Now I am become Death, the destroyer of worlds.”

Oppenheimer was the reluctant prince, not Krishna. “He didn’t want to kill people,” Wellerstein says. “But he knew that nuclear weapons were going to be built anyway, and he felt that he had a duty to do this horrible thing.”

5. Working Principle of Nuclear and Hydrogen Bombs

5.1. Fusion and Fission

All matter is composed of atoms: incredibly small structures that house different combinations of three particles, known as protons, neutrons, and electrons.

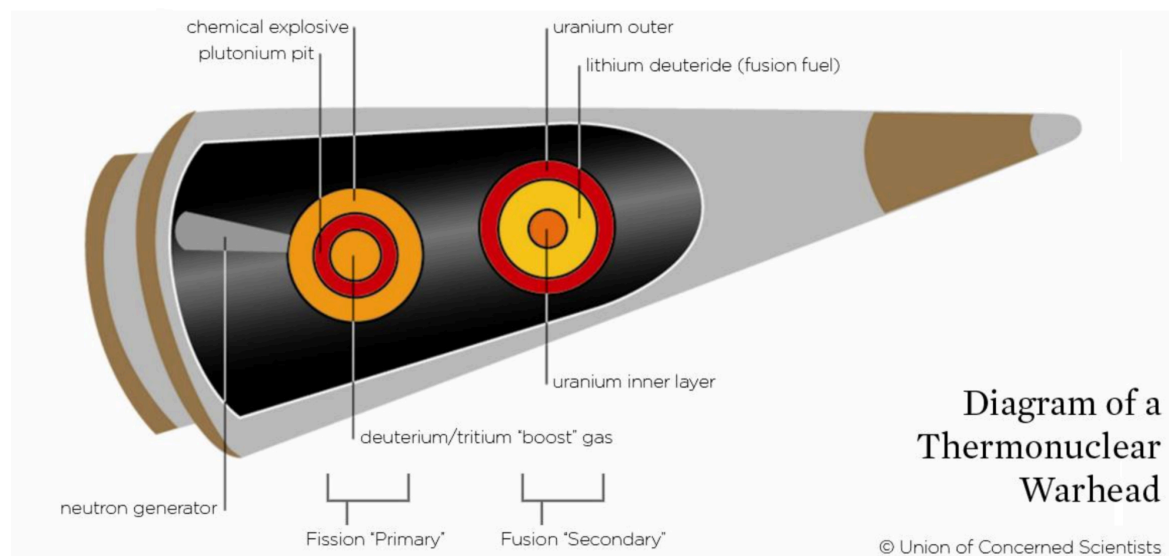
At the center of each atom is a “nucleus” (the plural of which is “nuclei”), where neutrons and protons are bound in proximity together. Most nuclei are relatively stable, meaning the makeup of their neutrons and protons is comparatively static and unchanging. During fission, the nuclei of certain heavy atoms split into smaller, lighter nuclei, releasing excess energy in the process. This can sometimes occur spontaneously, but can also, in certain nuclei, be induced from outside. If an unstable nucleus is hit by an outside neutron, it can cause the nucleus to split apart, or “fission.” In some elements the fission process also releases even more neutrons, which can trigger a chain reaction if they’re absorbed by nearby atoms. This is true for certain isotopes of uranium and plutonium. An isotope is a variation of the same element with different numbers of neutrons in the nucleus.

Fusion works in reverse: when exposed to extremely high temperatures and pressures, some lightweight nuclei can fuse to form heavier nuclei, releasing energy in the process.

In modern nuclear weapons, which use both fission and fusion, a single warhead can release more explosive energy in a fraction of a second than all of the weapons used during World War II *combined* including Fat Man and Little Boy, the two atom bombs dropped on Japan.

All nuclear weapons use fission to generate an explosion. “Little Boy” the first nuclear weapon ever used during wartime worked by shooting a hollow uranium 235 cylinder at a target “plug” of the same material.

Each piece by itself was not enough to constitute a critical mass (the minimum amount of nuclear material needed to maintain fission) but by colliding the pieces, critical mass was reached and a fission chain reaction occurred.



Modern nuclear weapons work slightly differently. Critical mass depends on the density of the material: as the density *increases*, the critical mass *decreases*. Instead of colliding two sub critical pieces of nuclear fuel, modern weapons detonate chemical

explosives around a sub critical sphere of uranium 235 or plutonium 239 metal. The force from the blast is directed inward, compressing the pit and bringing its atoms closer together. Once dense enough to reach the critical mass, neutrons are injected, initiating a fission chain reaction and producing an atomic explosion.

In fusion weapons (also called “thermonuclear” or “hydrogen” weapons), the energy from an initial fission explosion is used to “fuse” hydrogen isotopes together. The energy released by the weapon creates a fireball that reaches several tens of million degrees temperatures in the same range as the center of the sun (which also runs on fusion).

5.2 Required Materials and How Were They Found

Plutonium 239 is an isotope of plutonium. Plutonium 239 is the primary fissile isotope used for the production of nuclear weapons, although uranium 235 is also used for that purpose. Plutonium 239 is also one of the three isotopes that have been demonstrated to be usable as fuel in thermal spectrum nuclear reactors, along with uranium 235 and uranium 233. Plutonium 239 has a half life of 24,110 years.

The smaller critical mass of plutonium 239, as well as the ability to produce large amounts of nearly pure ^{239}Pu more cheaply than highly enriched weapons grade uranium 235, led to its use in nuclear weapons and nuclear power plants. A fission of an atom of uranium 235 (with 0.720% abundance) in a nuclear reactor produces two to three neutrons, and these neutrons can be absorbed by uranium 238 (with 99.3% abundance) to produce plutonium 239 and other isotopes. Plutonium-239 will also absorb neutrons and fission along with the uranium 235 in a reactor.

Of the common nuclear fuels, ^{239}Pu has the smallest critical mass. Its untampered spherical critical mass is about 11 kg (24.2 lbs), 10.2 cm in diameter. Using

appropriate triggers, neutron reflectors, implosion geometry, and tampers, the critical mass can be less than half of that.

The fission of one atom of ^{239}Pu generates $207.1 \text{ MeV} = 3.318 \times 10^{-11} \text{ J}$, i.e. $19.98 \text{ TJ/mol} = 83.61 \text{ TJ/kg}$, or about 23 gigawatt hours/kg.

Plutonium is made from the most abundant uranium isotope uranium 238. ^{239}Pu is normally created in nuclear reactors by transmutation of individual atoms of ^{238}U present in the fuel rods. Occasionally, when an atom of ^{238}U is exposed to neutron radiation, its nucleus will capture a neutron, changing it to ^{239}U . This happens more often with lower kinetic energy (as ^{238}U fission activation is 6.6 MeV). The ^{239}U then rapidly undergoes two β^- decays an emission of an electron and an anti-neutrino leaving a proton in the nucleus the first β^- decay transforming the ^{239}U into ^{239}Np , and the second β^- decay transforming the ^{239}Np into ^{239}Pu .

This reaction is relatively slow, so even after significant exposure, the ^{239}Pu is still mixed with a great deal of ^{238}U (and possibly other isotopes of uranium), oxygen, other components of the original material, and fission products. Only if the fuel has been exposed for a few days in the reactor can the ^{239}Pu be chemically separated from the rest of the material to yield high purity ^{239}Pu metal. ^{239}Pu has a higher probability for fission than ^{235}U and a larger number of neutrons produced per fission event, so it has a smaller critical mass. Pure ^{239}Pu also has a reasonably low rate of neutron emission due to spontaneous fission ($10 \text{ fission/s}\cdot\text{kg}$), making it feasible to assemble a mass that is highly supercritical before a detonation chain reaction begins.

In practice, however, reactor bred plutonium will invariably contain a certain amount of ^{240}Pu due to the tendency of ^{239}Pu to absorb an additional neutron during production. ^{240}Pu has a high rate of spontaneous fission events ($415,000 \text{ fission/s}\cdot\text{kg}$), making it an undesirable contaminant. As a result, plutonium containing a significant fraction of ^{240}Pu is not well suited to use in nuclear weapons; it emits neutron radiation, making handling more difficult, and its presence can lead to a "fizzle" in which a small explosion occurs, destroying the weapon but not causing fission of a significant fraction of the fuel. It is because of this limitation that

plutonium based weapons must be implosion type, rather than gun type. Moreover, ^{239}Pu and ^{240}Pu cannot be chemically distinguished, so expensive and difficult isotope separation would be necessary to separate them. Weapons grade plutonium is defined as containing no more than 7% ^{240}Pu ; this is achieved by only exposing ^{238}U to neutron sources for short periods of time to minimize the ^{240}Pu produced. Plutonium is classified according to the percentage of the contaminant plutonium-240 that it contains:

- * Supergrade 2–3%
- * Weapons grade 3–7%
- * Fuel grade 7–18%
- * Reactor grade 18% or more

A nuclear reactor that is used to produce plutonium for weapons, therefore, generally has a means for exposing ^{238}U to neutron radiation and for frequently replacing the irradiated ^{238}U with new ^{238}U . A reactor running on unenriched or moderately enriched uranium contains a great deal of ^{238}U . However, most commercial nuclear power reactor designs require the entire reactor to shut down, often for weeks, in order to change the fuel elements. They therefore produce plutonium in a mix of isotopes that is not well suited to weapon construction. Such a reactor could have machinery added that would permit ^{238}U slugs to be placed near the core and changed frequently, or it could be shut down frequently, so proliferation is a concern; for this reason, the International Atomic Energy Agency inspects licensed reactors often. A few commercial power reactor designs, such as the reaktor bolshoy moshchnosti kanalniy (RBMK) and pressurized heavy water reactor (PHWR), do permit refueling without shutdowns, and they may pose a proliferation risk. By contrast, the Canadian CANDU heavy water moderated, natural uranium fueled reactor can also be refueled while operating, but it normally consumes most of the ^{239}Pu it produces in situ; thus, it is not only inherently less proliferative than most reactors, but can even be operated as an "actinide incinerator". The American IFR (Integral Fast Reactor) can also be operated in an incineration mode, having some advantages in not accumulating the

plutonium 242 isotope or the long lived actinides, which cannot be easily burned except in a fast reactor. Also, IFR fuel has a high proportion of burnable isotopes, while in CANDU an inert material is needed to dilute the fuel; this means the IFR can burn a higher fraction of its fuel before needing reprocessing. Most plutonium is produced in research reactors or plutonium production reactors called breeder reactors because they produce more plutonium than they consume fuel. In principle, such reactors make extremely efficient use of natural uranium. In practice, their construction and operation are sufficiently difficult that they are generally only used to produce plutonium. Breeder reactors are generally (but not always) fast reactors, since fast neutrons are somewhat more efficient at plutonium production.

Nuclide [n 1]	Z	N	Isotopic mass (Da) ^[4] [n 2][n 3]	Discovery year ^{[5][6]}	Half-life ^[1]	Decay mode ^[1] [n 4]	Daughter isotope [n 5][n 6]	Spin and parity ^[1] [n 7][n 8]	Isotopic abundance
²²⁶ Pu ^[7]	94	132	226.03825(22)#	(2024) 	≥1 ms	α	²²² U	0+	
²²⁷ Pu ^[8]	94	133	227.03947(11)#	2024 	0.78 ^{+0.39} _{−0.19} s	α	²²³ U	5/2+#	
²²⁸ Pu	94	134	228.038763(25)	1994 	2.1(13) s	α	²²⁴ U	0+	
²²⁹ Pu	94	135	229.040145(65)	1994 	91(26) s	α (~50%)	²²⁵ U	3/2+#	
						β ⁺ (~50%)	²²⁹ Np		
						SF (<7%)	(various)		
²³⁰ Pu	94	136	230.039648(16)	1990 	105(10) s	α (>73%) ^[9]	²²⁶ U	0+	
						β ⁺ (<27%)	²³⁰ Np		
²³¹ Pu	94	137	231.041126(24)	1999 	8.6(5) min	β ⁺ (87%)	²³¹ Np	(3/2+)	
						α (13%)	²²⁷ U		
²³² Pu	94	138	232.041182(18)	1973 	33.7(5) min	EC (77%) ^[10]	²³² Np	0+	
						α (23%)	²²⁸ U		
²³³ Pu	94	139	233.042997(58)	1957 	20.9(4) min	β ⁺ (99.88%)	²³³ Np	5/2+#	
						α (0.12%)	²²⁹ U		
²³⁴ Pu	94	140	234.0433175(73)	1949 	8.8(1) h	EC (94%)	²³⁴ Np	0+	
						α (6%)	²³⁰ U		
²³⁵ Pu	94	141	235.045285(22)	1957 	25.3(5) min	β ⁺	²³⁵ Np	(5/2+)	
						α (0.0028%)	²³¹ U		
²³⁶ Pu	94	142	236.0460567(19)	1949 	2.858(8) y	α ^[n 9]	²³² U	0+	
						SF (1.9 × 10 ^{−7} %)	(various)		
						CD (2 × 10 ^{−12} %)	²⁰⁸ Pb ²⁸ Mg		
^{236m} Pu			1185.45(15) keV	2005 	1.2(3) μs	IT	²³⁶ Pu	5−	
²³⁷ Pu	94	143	237.0484079(18)	1949 	45.64(4) d	EC	²³⁷ Np	7/2−	
						α (0.0042%)	²³³ U		
^{237m1} Pu			145.543(8) keV	1957 	180(20) ms	IT	²³⁷ Pu	1/2+	
^{237m2} Pu			2600(20) keV	1969 	103(7) ns	SF	(various)		
^{237m3} Pu			2900(250) keV	1970 	1.1(1) μs	SF	(various)		

Plutonium 239 is more frequently used in nuclear weapons than uranium 235, as it is easier to obtain in quantity capable of criticality. The process of enriching uranium, i.e., increasing the ratio of ²³⁵U to ²³⁸U to weapons grade, is generally a more lengthy and costly process than the production of plutonium 239 from ²³⁸U and subsequent chemical separation.

Plutonium (94Pu) is an artificial element, except for trace quantities resulting from neutron capture by uranium, and thus a standard atomic weight cannot be given. Like all artificial elements, it has no stable isotopes. It was synthesized before being found in nature, with the first isotope synthesized being ²³⁸Pu in 1940. Twenty two plutonium radioisotopes have been characterized. The most stable are ²⁴⁴Pu with a

half life of 81.3 million years, ^{242}Pu with a half life of 375,000 years, ^{239}Pu with a half life of 24,110 years, and ^{240}Pu with a half life of 6,561 years. This element also has eight meta states; all have half lives of less than one second. The known isotopes of plutonium range from ^{226}Pu to ^{247}Pu . The primary decay modes before the most stable isotope, ^{244}Pu , are spontaneous fission and alpha decay; the primary mode after is beta emission. The primary decay products before ^{244}Pu are isotopes of uranium and neptunium, and the primary decay products after are isotopes of americium.

Uranium 235 (^{235}U or U-235) is an isotope of uranium making up about 0.72% of natural uranium. Unlike the predominant isotope uranium 238, it is fissile, i.e., it can sustain a nuclear chain reaction. It is the only fissile isotope that exists in nature as a primordial nuclide and was discovered in 1935 by Arthur Jeffrey Dempster. The release of energy from the fission of uranium 235 powers most nuclear reactors and nuclear weapons. Uranium enrichment is the process of taking natural uranium and increasing the proportion of uranium 235. Some reactor types can use natural uranium directly, while nuclear weapons and some others reactor types require uranium enriched in U 235.

Uranium 235 has a half life of 704 million years. Its fission cross section for slow thermal neutrons is about 584.3 ± 1 barns. For fast neutrons, it is on the order of 1 barn. Most neutron absorptions induce fission, though a minority (about 15%) result in the formation of uranium 236.

The fission of one atom of uranium 235 releases 202.5 MeV (3.24×10^{-11} J) inside the reactor. That corresponds to 19.54 TJ/mol, or 83.14 TJ/kg. Another 8.8 MeV escapes the reactor as anti-neutrinos. When $^{235}_{92}\text{U}$ nuclei are bombarded with neutrons, one of the many fission reactions that they can undergo is the following:

1

$0n + ^{235}\text{U}$

$\rightarrow ^{236}\text{U}$

$\rightarrow ^{144}\text{U}$

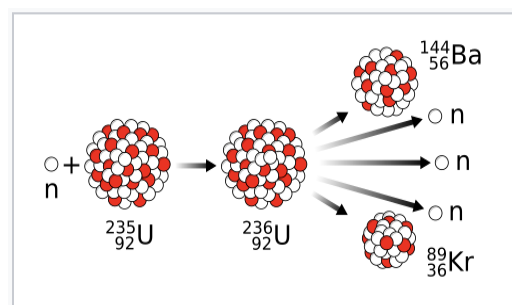
$^{56}\text{Ba} + 89$

$^{36}\text{Kr} + 31$

$0n$

Heavy water reactors and some graphite moderated reactors can use natural uranium, but light water reactors must use low enriched uranium because of the higher neutron absorption of light water. Uranium enrichment removes some of the uranium 238 and increases the proportion of uranium 235. Highly enriched uranium (HEU), which contains an even greater proportion of uranium 235, is sometimes used in the reactors of nuclear submarines, research reactors and nuclear weapons.

If at least one neutron from uranium 235 fission strikes another nucleus and causes it to fission, then the chain reaction will continue. If the reaction continues to sustain itself, it is said to be critical, and the mass of ^{235}U required to produce the critical condition is said to be a critical mass. A critical chain reaction can be achieved at low concentrations of ^{235}U if the neutrons from fission are moderated to lower their speed, since the probability for fission with slow neutrons is greater. A fission chain reaction produces intermediate mass fragments which are highly radioactive and produce further energy by their radioactive decay. Some of them produce neutrons, called delayed neutrons, which contribute to the fission chain reaction. The power output of nuclear reactors is adjusted by the location of control rods containing elements that strongly absorb neutrons, e.g., boron, cadmium, or hafnium, in the reactor core. In nuclear bombs, the reaction is uncontrolled and the large amount of

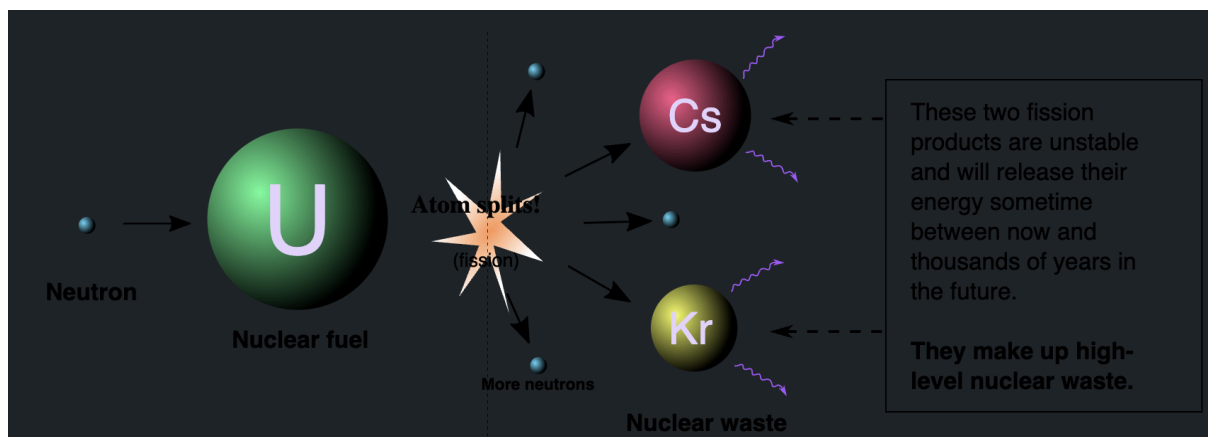


Nuclear fission seen with a uranium-
 ^{235}U nucleus

energy released creates a nuclear explosion.

5.3 Nuclear Waste and General Problems during construction and Research

Nuclear energy is released when a nuclear fuel nucleus snaps into two in a reactor. The key component of nuclear waste is the leftover smaller nuclei, known as fission products.

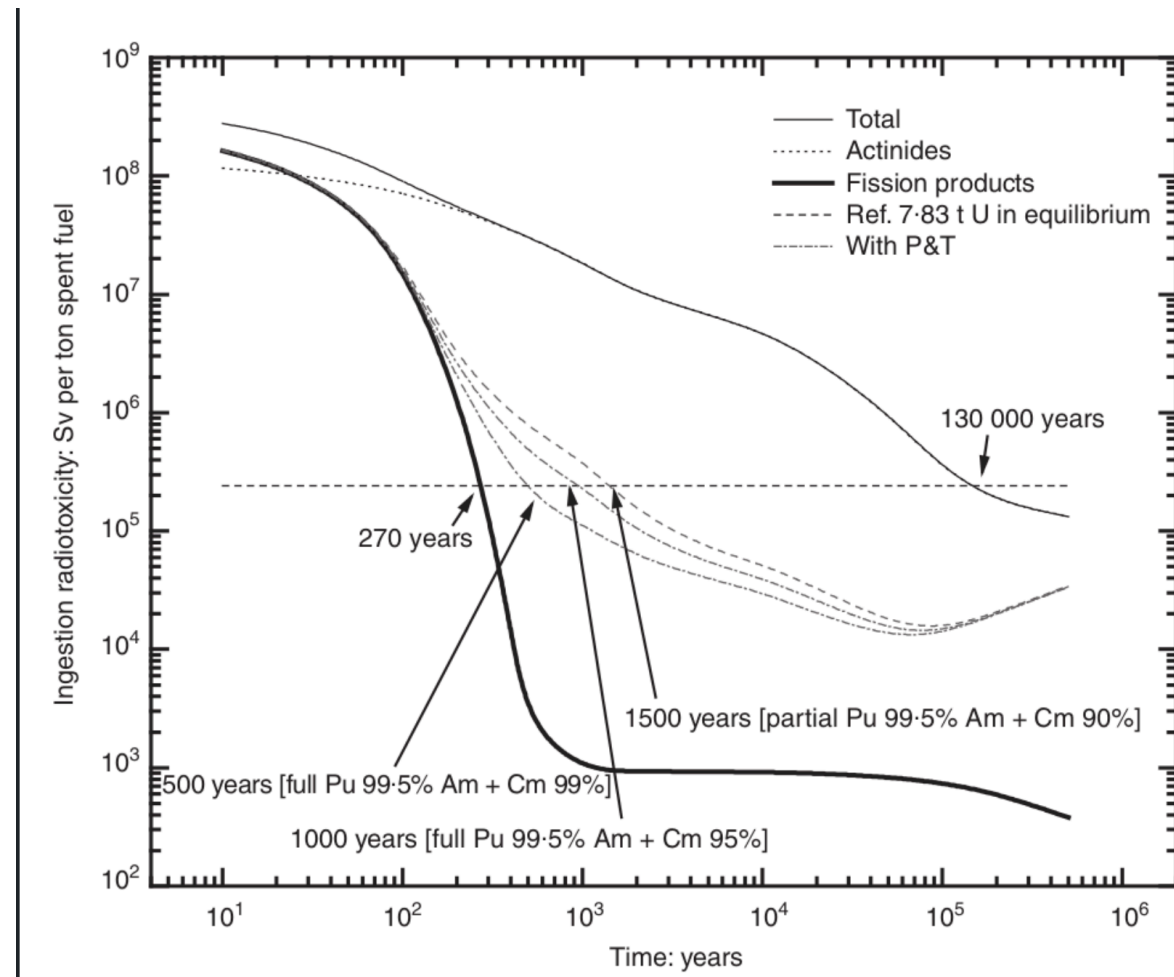


From the outside, nuclear waste looks exactly like the fuel that was loaded into the reactor typically assemblies of cylindrical metal rods enclosing fuel pellets. But because nuclear reactions have occurred, the contents aren't quite the same.

Like everything, nuclear waste is made of elements you can find on the periodic table, including isotopes of iron, zinc, germanium, zirconium, silver, and iodine. You never know which two fission product elements you're going to get for a given fission event, but you always get the same average composition as billions upon billions of atoms split. You basically get a huge variety of elements, shown below.

[illegible]

The nuclear waste question fundamentally asks whether or not humanity can prevent this radioactive material from causing harm to people and the environment. This question was recognized early on in nuclear history, by Enrico Fermi himself (the lead scientist in charge of the world's first man made nuclear chain reaction), who stated:



It is not clear that the public will accept an energy source that produces this much radioactivity

— Enrico Fermi

6. Historical Background

Building on major scientific breakthroughs made during the 1930s, the United Kingdom began the world's first nuclear weapons research project, codenamed Tube Alloys, in 1941, during World War II. The United States, in collaboration with the United Kingdom, initiated the Manhattan Project the following year to build a weapon using nuclear fission. The project also involved Canada. In August 1945, the atomic bombings of Hiroshima and Nagasaki were conducted by the United States, with British consent, against Japan at the close of that war, standing to date as the only use of nuclear weapons in hostilities.

The Soviet Union started development shortly after with their own atomic bomb project, and not long after, both countries were developing even more powerful fusion weapons, known as hydrogen bombs. Britain and France built their own systems in the 1950s, and the number of states with nuclear capabilities has gradually grown larger in the decades since.

A nuclear weapon, also known as an atomic bomb, possesses enormous destructive power from nuclear fission, or a combination of fission and fusion reactions.

In the first decades of the 19th century, physics was revolutionized with developments in the understanding of the nature of atoms, including the discoveries in atomic theory by John Dalton. Around the turn of the 20th century, it was discovered by Hans Geiger and Ernest Marsden and then Ernest Rutherford that atoms had a highly dense, very small, charged central core called an atomic nucleus. In 1898, Pierre and Marie Curie discovered that pitchblende, an ore of uranium, contained a substance which they named radium that emitted large amounts of radiation. Ernest Rutherford and Frederick Soddy identified that atoms were breaking down and turning into different elements. Hopes were raised among scientists and laymen that the elements around us could contain tremendous amounts of unseen energy, waiting to be harnessed. In 1905, Albert Einstein described this potential in his famous equation, $E = mc^2$.

H. G. Wells was inspired by the work of Rutherford to write about an "atom bomb" in a 1914 novel, *The World Set Free*, which appeared shortly before the First World War. In a 1924 article, Winston Churchill speculated about the possible military implications: "Might not a bomb no bigger than an orange be found to possess a secret power to destroy a whole block of buildings nay to concentrate the force of a thousand tons of cordite and blast a township at a stroke?"

At the time however, there was no known mechanism which could be used to unlock the vast energy potential that was theorized to exist inside the atom. The only particle then known to exist within the nucleus was the positively charged proton, which would act to repel protons set in motion towards it. Then in 1932, a key breakthrough was made with the discovery of the neutron. Having no electric charge, the neutron is able to penetrate the nucleus with relative ease.

In January 1933, the Nazis came to power in Germany and suppressed Jewish scientists. Physicist Leo Szilard fled to London where, in 1934, he patented the idea of a nuclear chain reaction using neutrons. The patent also introduced the term critical mass to describe the minimum amount of material required to sustain the chain reaction and its potential to cause an explosion. The patent was not about an atomic bomb per se, as the possibility of chain reaction was still very speculative. Szilard subsequently assigned the patent to the British Admiralty so that it could be covered by the Official Secrets Act. This work of Szilard's was ahead of the time, five years before the public discovery of nuclear fission and eight years before a working nuclear reactor. When he coined the term neutron induced chain reaction, he was not sure about the use of isotopes or standard forms of elements. Despite this uncertainty, he correctly theorized uranium and thorium as primary candidates for such a reaction, along with beryllium which was later determined to be unnecessary in practice. Szilard joined Enrico Fermi in developing the first uranium-fuelled nuclear reactor, Chicago Pile-1, which was activated at the University of Chicago in 1942.

In Paris in 1934, Irène and Frédéric Joliot Curie discovered that artificial radioactivity could be induced in stable elements by bombarding them with alpha particles; in Italy Enrico Fermi reported similar results when bombarding uranium with neutrons. He mistakenly believed he had discovered elements 93 and 94, naming them ausenium and hesperium. In 1938 it was realized these were in fact fission products. In December 1938, Otto Hahn and Fritz Strassmann reported that they had detected the element barium after bombarding uranium with neutrons. Lise Meitner and Otto Robert Frisch correctly interpreted these results as being due to the splitting of the uranium atom. Frisch confirmed this experimentally on January 13, 1939. They gave the process the name "fission" because of its similarity to the splitting of a cell into two new cells. Even before it was published, news of Meitner's and Frisch's interpretation crossed the Atlantic. In their second publication on nuclear fission in February 1939, Hahn and Strassmann predicted the existence and liberation of additional neutrons during the fission process, opening up the possibility of a nuclear chain reaction. After learning about the German fission in 1939, Leo Szilard concluded that uranium would be the element which could realize his 1933 idea about nuclear chain reaction.

In the United States, scientists at Columbia University in New York City decided to replicate the experiment and on January 25, 1939, conducted the first nuclear fission experiment in the United States in the basement of Pupin Hall. The following year, they identified the active component of uranium as being the rare isotope uranium 235.

Between 1939 and 1940, Joliot-Curie's team applied for a patent family covering different use cases of atomic energy, one being the first official document explicitly mentioning a nuclear explosion as a purpose, including for war. This patent was applied for on May 4, 1939, but only granted in 1950, being withheld by French authorities in the meantime. Uranium appears in nature primarily in two isotopes: uranium 238 and uranium 235. When the nucleus of uranium 235 absorbs a neutron, it undergoes nuclear fission, releasing energy and, on average, 2.5 neutrons. Because

uranium 235 releases more neutrons than it absorbs, it can support a chain reaction and so is described as fissile. Uranium 238, on the other hand, is not fissile as it does not normally undergo fission when it absorbs a neutron.

By the start of the war in September 1939, many scientists likely to be persecuted by the Nazis had already escaped. Physicists on both sides were well aware of the possibility of utilizing nuclear fission as a weapon, but no one was quite sure how it could be engineered. In August 1939, concerned that Germany might have its own project to develop fission based weapons, Albert Einstein signed a letter to U.S. President Franklin D. Roosevelt warning him of the threat.

Roosevelt responded by setting up the Uranium Committee under Lyman James Briggs but, with little initial funding (\$6,000), progress was slow. It was not until the U.S. entered the war in December 1941 that Washington decided to commit the necessary resources to a top secret high priority bomb project. Organized research first began in Britain as part of the Tube Alloys project, the world's first nuclear weapons project, which also involved Canada. The Maud Committee was set up following the work of Frisch and Rudolf Peierls who calculated uranium 235's critical mass and found it to be much smaller than previously thought which meant that a deliverable bomb should be possible. In the February 1940 Frisch Peierls memorandum they stated that: "The energy liberated in the explosion of such a super bomb...will, for an instant, produce a temperature comparable to that of the interior of the sun. The blast from such an explosion would destroy life in a wide area. The size of this area is difficult to estimate, but it will probably cover the centre of a big city."

Edgar Sengier, a director of Shinkolobwe Mine in the Congo which produced by far the highest quality uranium ore in the world, had become aware of uranium's possible use in a bomb. In late 1940, fearing that it might be seized by the Germans, he shipped the mine's entire stockpile of ore to a warehouse in New York. For 18 months British research outpaced the American but by mid 1942, it became apparent that the industrial effort required was beyond Britain's already stretched wartime economy.

In September 1942, General Leslie Groves was appointed to lead the U.S. project which became known as the Manhattan Project. Two of his first acts were to obtain authorization to assign the highest priority AAA rating on necessary procurements, and to order the purchase of all 1,250 tons of the Shinkolobwe ore. The Tube Alloys project was quickly overtaken by the U.S. effort and after Roosevelt and Churchill signed the Quebec Agreement in 1943, it was relocated and amalgamated into the Manhattan Project. Canada provided uranium and plutonium for the project.

Szilard started to acquire high quality graphite and uranium, which were the necessary materials for building a large scale chain reaction experiment. The Metallurgical Laboratory at the University of Chicago was tasked with the completion of such a reactor, and Fermi moved there, continuing the pile experiments he began at Columbia. After many subcritical designs, Chicago Pile 1 achieved criticality on December 2, 1942. The success of this demonstration and technological breakthrough were partially due to Szilard's new atomic theories, his uranium lattice design, and the identification and mitigation of a key graphite impurity (boron) through a joint collaboration with graphite suppliers.

7. The Manhattan Project

The Manhattan Project was a research and development program undertaken during World War II to produce the first nuclear weapons. It was led by the United States in collaboration with the United Kingdom and Canada. The Manhattan Project employed nearly 130,000 people at its peak and cost nearly US\$2 billion (equivalent to about \$28 billion in 2024).

From 1942 to 1946, the project was directed by Major General Leslie Groves of the U.S. Army Corps of Engineers. Nuclear physicist J. Robert Oppenheimer was the director of the Los Alamos Laboratory that designed the bombs. The Army program was designated the Manhattan District, as its first headquarters were in Manhattan; the

name superseded the initial codename, Development of Substitute Materials, for the entire project. The project absorbed its earlier British counterpart, Tube Alloys.

Over 80 percent of the project cost was for building and operating the fissile material production plants. The project proposed both highly enriched uranium and plutonium as fuel for nuclear weapons. Enriched uranium was produced at the Clinton Engineer Works in Tennessee. Plutonium was produced in the world's first industrial scale nuclear reactors at the Hanford Engineer Works in Washington. These sites were supported by dozens of other facilities across Canada, the US and the UK.

The weapons designs were produced at the Los Alamos Laboratory (Project Y) in New Mexico and resulted in two weapon designs used during the war: Little Boy (enriched uranium gun type) and Fat Man (plutonium implosion). The Fat Man design was initially a low priority fallback option, as it was complex and required explosive lenses, but in 1944 it was confirmed that plutonium from Hanford was not suitable for a gun-type bomb because of the high proportion of plutonium 240. The first nuclear device ever detonated was an implosion type bomb during the Trinity test, conducted at White Sands Proving Ground in New Mexico on 16 July 1945. The project was responsible for developing the specific means of delivering the weapons onto military targets, and for the use of the Little Boy and Fat Man bombs in the atomic bombings of Hiroshima and Nagasaki in August 1945.

The project was also charged with gathering intelligence on the German nuclear weapon project. Through Operation Alsos, Manhattan Project personnel served in Europe, sometimes behind enemy lines, where they gathered nuclear materials and documents and rounded up German scientists. Despite the Manhattan Project's own emphasis on security, Soviet atomic spies penetrated the program.

In the immediate postwar years, the Manhattan Project conducted weapons testing at Bikini Atoll as part of Operation Crossroads, developed new weapons, promoted the development of the network of national laboratories, supported medical research into

radiology, and laid the foundations for the nuclear navy. It maintained control over American atomic weapons research and production until the formation of the United States Atomic Energy Commission (AEC) in January 1947.

7.1. The Headquarters, Building Process and Selecting Participants

Oak Ridge, Tennessee

On September 19, 1942, two days after being assigned the lead role in developing the Manhattan Project, General Groves approved an idyllic location in East Tennessee, a series of valleys and rolling hills approximately twenty miles west of Knoxville. With the Appalachian mountains to the east and the Cumberland Plateau to the west, this sparsely populated area proved to be an ideal choice to develop what would eventually become Oak Ridge. At the time of selection, Oak Ridge was named the Kingston Demolition Range and later the Clinton Engineer Works. Oak Ridge was slated to hold four production plants: gaseous diffusion, pile, centrifuge, and electromagnetic. Prior to selecting this site, General Groves' team scouted several locations throughout the US. Chicago was considered but was deemed too densely populated for the scale of development required. California was also scouted but was too close to the coast and therefore too vulnerable to potential enemy attack. Lastly, several sites in Washington state were considered but too much work was required to install power lines for the immense energy needed and, like California, it was considered vulnerable to potential attack.

East Tennessee fit the bill perfectly. The ridges and valleys allowed for natural separation of the proposed plants. The sparse population meant that in the event of a catastrophe any potential loss of life would be low. The site was far enough inland to prevent a potential enemy attack. In addition, the various nearby hydroelectric dams operated by the Tennessee Valley Authority (TVA) ensured that the site could be

efficiently powered. Lastly, displacement of residents was deemed to be more manageable than in a more urban location. The immediate area had only about 1,000 families at the time.

During the fall of 1942, land acquisition began with the purchase of over 55,000 acres at a cost of approximately \$3.5 million in 1942 dollars. By spring of 1943 the site was officially designated Clinton Engineer Works, and in the fall of that year, construction had begun on a massive building for enriching uranium. The need for construction workers at this new gaseous diffusion plant, named K-25, and other uranium enrichment facilities in the city caused the population of Oak Ridge to increase from 13,000 people as originally planned to over 75,000 by the end of the war. At its peak, K-25 employed 12,000 workers and was one of the linchpins of what would become the most populous Manhattan Project site in the country.

Hanford, Washington

While the Clinton Engineer Works (Oak Ridge) was developed by Groves and his team to enrich uranium, a site also had to be found to produce plutonium. With Knoxville, Tennessee being so close to Oak Ridge, Groves ruled out the addition of a plutonium production nuclear reactor in the region. A potential failure of a reactor could have dire consequences for the population of the nearby city. Once again, Groves turned to Washington state as an option.

Any potential plutonium production site required three things: water, power, and a large area with few people. The Pacific Northwest met the criteria. The large and fast-moving Columbia River could provide water for cooling the reactors, and the Grand Coulee Dam on the Columbia could provide the hydroelectric power needed. The lightly populated open expanse of south eastern Washington could provide the space to ensure safe and efficient operation. In January of 1943, after having personally visited the site, General Groves acquired 780 square miles of land along the Columbia at a cost of \$5.1 million in 1943 dollars. With sufficient hydroelectric dams

and power lines, the Hanford Engineer Works began construction, so named because of a tiny village of less than 100 residents located within the plot of land. The Manhattan Project now had an ideal location for plutonium production. By October 1943, construction of Hanford's B Reactor had begun.

7.2. Inside and Outside Threats

7.2.1. Document And Knowledge Leaks

Efforts to keep a lid on atomic energy research had begun even before the Manhattan Project had started. In 1939, refugee scientists—mindful of the growing threat posed by Nazi Germany—urged their U.S. colleagues to undertake voluntary censorship of published studies concerning uranium fission. Initially, they rejected the idea—but as one European country after another fell to the Germans in 1940, the U.S. scientific community came to understand the importance of self-censorship. The Division of Physical Sciences of the National Research Council established a committee that succeeded in convincing most scientists to withhold publication of papers on sensitive subjects.

Still, occasional newspaper articles wrote about the potential applications of uranium fission in warfare. One such article, sympathetic to the plight of Europe, appeared in the March 18, 1941 edition of the Montana Standard:

It is known that there is enough power in one portion of uranium-235 the size of a pear to shoot the Empire State Building in the air with the speed of a rocket. It would be more than two million times as possible as one 1,000-pound air bomb of the type used by the Nazis in bombing London.

But nature has used her usual foresight in guarding such potential annihilation. Scientists admit it is impossible to separate uranium-235 from uranium-238. It is not that science has not figured out a method. It has. But the apparatus would be so

complicated there is not now in existence enough scientists to set it up, even though they were to work a lifetime doing it.

We might say, quite frankly, that our scientists have disillusioned us.

Although physicists continued to play up the “impossible task” of separating sufficient amounts of uranium-235 to construct a bomb, reports and rumors of such a secret weapon continued to pop up in the press. Finally, in 1943, the director of the wartime Office of Censorship sent a special request to editors and broadcasters to not report on any rumors of “secret weapons” and to withhold any information on:

Production or utilization of atom smashing, atomic energy, atomic fission, atomic splitting, or any of their equivalents.

The use for military purposes of radium or radioactive materials, heavy water, high voltage discharge equipment, cyclotrons.

The following elements or any of their compounds: plutonium, uranium, ytterbium, hafnium, protoactinium, radium, thorium, deuterium.

The Manhattan Project to develop the first atomic bomb during World War II was among the most highly classified and tightly secured programs ever undertaken by the U.S. government. Nevertheless, it generated more than 1,500 leak investigations involving unauthorized disclosures of classified Project information.

In several respects, the Manhattan Project established the template for secret government programs during the Cold War (and after). It pioneered or refined the practices of compartmentalization of information, “black” budgets, cover and deception to conceal secret facilities, minimal notification to Congress, and more.

But wherever there are national security secrets, it seems that leaks and spies are not far behind. During the course of the Manhattan Project, counterintelligence agents “handled more than 1,000 general subversive investigations, over 1,500 cases in which classified project information was transmitted to unauthorized persons, approximately 100 suspected espionage cases, and approximately 200 suspected sabotage cases,” according to the newly declassified [history](#) (at pp. S2-3).

Since September 1943, investigations were conducted of more than 1500 'loose talk' or leakage of information cases and corrective action was taken in more than 1200 violations of procedures for handling classified material. The declassified official history of the Manhattan Project is fascinating in that it reveals the tremendous amount of resources invested in tracking down the source of each and every reported leak.

Upon discovery of the source of a violation of regulations for safeguarding military information, the violator, if a project employee, was usually reprimanded, informed of the possible application of the Espionage Act, and warned not to repeat the violation."

Here's a sampling of some of the cases of leaks discussed in official documents of the project:

1) A Naval lieutenant at a dinner party in March 1945 openly wondered "when we would use the atomic bomb," adding that "only about ten pounds of U-235 are needed to end the war." He also said that "several thousand persons were working on it in Tennessee."

When agents interviewed the lieutenant, he recalled a 1942 lecture given at the chemistry department at Harvard regarding the possibilities of atomic power, and he had since discussed the subject with friends and had read many speculative articles in technical magazines. He stated that "anyone traveling for the government realized there was a great secret project there" and he merely assumed that it was engaged in the development of atomic power.

2) In the fall of 1944, an employee of a New York firm, doing engineering work for the project, was taking a train from New York to Oak Ridge carrying a highly secret file of engineering details. He arrived at Penn Station with several minutes to spare before the train departed, so he used the time to make a last minute call to his wife from a telephone booth.

After finishing the call, he headed for the train and discovered that he didn't have the secret papers with him. He decided to catch the train, so he called his office and requested that someone come down to the station to look for the papers. The security agent for the company rushed to the station and made a frantic search, finally locating them at the information booth. The employee who had lost the papers was ordered to return to New York, where he was "soundly reprimanded" and fired by the company.

3) One case concerned a pamphlet titled Startling Power. In the winter of 1945, a patent engineer secured a copy of the pamphlet, which had been published by the Good News Publishing Company in Chicago and was distributed by the Moody Bible Institute. The pamphlet declared that, "Uranium 235, extracted from natural uranium ore, promises to make all of our power sources mere child's toys by comparison. Professor John R. Dunning, Columbia University Leader in Atomic Energy Research, has stated that the natural substances from which Uranium 285 is extracted abound abundantly in the earth and throughout many sections of the world." However, the pamphlet continued, "We must not overlook the far more vital and assured fact that God has given to Christians the gift of the Holy Spirit with energies far more dynamic than those of exploding atoms or mysterious elements."

Upon reading this pamphlet, the patent engineer concluded that U-235 merited investigation by his company. His boss contacted the prominent physicist Arthur Compton to find out more about the potential applications of uranium, which triggered an investigation. Security officials were relieved to learn the source of the inquiry was a harmless reading of a religious tract.

Religious sermons were occasionally investigated over the years, such as when a Lutheran minister declared to his flock that, "One of the developments in the field of science today is a new source of energy called uranium-235. But regardless of the power of uranium 235 or other energy which science may discover, it will never be powerful enough to comfort us in affliction or strengthen us in despair. We must search out the Lord for those things." Discrete inquires revealed that this, and other sermons like it, were taken from the same pamphlet, Startling Power.

Fundamentally, however, information security was not to be achieved by the force of law or the threat of punishment. Rather, it was rooted in shared values and common commitments, [the Project history](#) said.

“Grounds for protecting information were largely patriotism, loyalty to the fighting men, and the reasoning that the less publicity given the Project, the more difficult it would be for the enemy to acquire information about it and also, the greater would be the element of surprise” (p. 6.13).

7.2.2. Spies

Considering the existing tense environment in the world at that time, All states had deployed spies in others’ military related projects to gain information and to not fall behind in the arms race. This applied critically to the Manhattan Project as well hence there were several cases of “atomic spies” leaking information regarding the project itself to other states or unwanted parties.

One of such spies working for the Soviets was John Cairncross. John Cairncross (1913-1995) was a British literary scholar, civil servant, and Soviet atomic spy. In the 1990s, Cairncross was identified as the “fifth man” in the Cambridge spy ring, which consisted of four other Soviet double agents: Anthony Blunt, Guy Burgess, Donald Maclean, and Kim Philby.

After joining the Foreign Service in 1936, he was recruited by James Klugmann, an influential communist at Cambridge and Soviet liaison, to become a Soviet spy. Following his move to the Treasury in 1938, he transferred once more in 1940 to the Cabinet Office, where he became the private secretary of Sir Maurice Hankey, the Chancellor of the Duchy of Lancaster.

In his oral history on the Atomic Heritage Foundation's [*Voices of the Manhattan Project*](#) website, historian John Earl Haynes notes that Sir Hankey was also a part of the committee that supervised the British atomic program. According to Haynes, this would have given Cairncross "access to all the reports that were coming to Hankey's office" and allow him to "pass them on to the Soviets."

Besides possibly passing on information from the British scientists' MAUD Report, Cairncross also provided Moscow with the list of American scientists, who came to England "to meet with the senior scientists of the British atomic program." This list of names would later be given to the American NKVD (forerunner to the KGB) station in New York, in order to try to recruit these individuals to become Soviet agents.

Another spy of the Soviets was a British Secretary named Melita Norwood who was also the Soviet Union's longest serving spy which according to her testimony in 1999 she has been a spy for nearly four decades. It's unclear how much Norwood's espionage helped the Soviet atomic program, but she [was officially honored](#) for her work when she visited Moscow in 1979. Finally exposed as a spy in the 1990s, Norwood "cheerfully admitted what she had done, and said she would do it again," says Harvey Klehr, emeritus professor of politics and history at Emory University and author of various books on Soviet espionage.

Another spy was Klaus Fuchs, who was a German theoretical physicist and spy who worked at Los Alamos during the Manhattan Project and passed atomic secrets to the Soviet Union.

Fuchs was transferred to Los Alamos in August 1944, where he worked in the Theoretical Division under Hans Bethe and Edward Teller. There, he calculated the approximate energy yield of an atomic explosion, and specialized in researching implosion methods, focusing in particular on the "Fat Man" implosion bomb. Additionally, he was present at the Trinity Test on July 16, 1945.

Fuchs continued to spy on the Anglo-American atomic bomb project for the Soviet Union while at Los Alamos. His primary point of contact was Harry Gold, who served as a courier for a number of other spies at Los Alamos.

In addition to providing Gold with secrets on the American atomic project, Fuchs also passed detailed information about the hydrogen bomb to the Soviet Union. Some experts estimate that Fuchs' intelligence enabled the Soviets to develop and test their own atomic bomb one to two years earlier than otherwise expected.

Another spy was David Greenglass, who was a machinist and member of the Special Engineer Detachment who engaged in espionage activities for the Soviet Union at Los Alamos during the Manhattan Project.

Greenglass was assigned to work on the Manhattan Project in July 1944 as a member of the Special Engineer Detachment. Initially stationed at Oak Ridge, Tennessee, he was transferred to Los Alamos in August 1944 where he worked until February 1946. An Army sergeant and skilled machinist, Greenglass was on a team tasked with making molds for the high-explosive lenses used to detonate the plutonium core in the implosion bomb.

In November 1944, Greenglass and his wife were recruited by his brother-in-law, Julius Rosenberg, to spy for the Soviet Union. Decrypted cables from the United States Army Signal Intelligence Service (SIS)'s "Venona" project indicate Greenglass and his wife were given the codenames KALIBR (Caliber), and OSA (Wasp), respectively. In the middle of 1945, Greenglass sent Rosenberg a crude sketch and twelve pages of detailed notes on the implosion-type bomb.

There were many other spies for many sides in the Manhattan Project. They all had different impacts, they all leaked different aspects of the project and all separate leaks had to be taken into account to actually stay ahead during the war.

7.3. Safety Measures

7.3.1. For the Nuclear Testing and General Radioactivity

Protecting workers was an important priority for Manhattan Project officials. The radically new hazards of atomic energy presented significant challenges for project directors who were responsible for worker safety. Once the District went into production, it was an extremely dangerous place to work and live. Workers were handling a variety of toxic chemicals, using hazardous liquids and gases under high pressures, working with high voltages, and performing experiments involving explosives, not to mention the largely unknown dangers presented by radioactivity and handling fissile materials. To address these challenges, the Manhattan Project imposed strict safety measures in production facilities and created an entire medical and health section to study the effects of radiation on workers, including radiation counters and reactor safety techniques. In December 1945, the National Safety Council presented the Manhattan Project with the Award of Honor for Distinguished Service to Safety in recognition of its safety record. Between January 1943 and June 1945, there were 62 fatalities and 3,879 disabling injuries, which was about 62 percent below that of private industry.

On November 3, 1943, General Leslie Groves appointed Stafford Warren as chief of the MED's medical section and medical advisor to the Manhattan Project. Warren's initial task was to staff hospitals at Oak Ridge, Hanford, and Los Alamos to provide adequate medical care for each site's rapidly growing workforce. After each site became fully operational, Warren's medical section focused their efforts on medical research and was responsible for the MED's health and safety programs. Warren would reiterate, however, that medicine was subservient to the larger goal of "winning

the present war” and presented the medical research program as funded and directed to “strengthen government interests from a medico-legal point of view”.

To Warren and other MED officials, protecting workers from atomic energy hazards was important but did not merit unusual or excessive safety standards. He believed that the atomic weapons industry should be treated as simply another industrial program under which “normal industrial practices were good enough” and that special standards should apply only upon proof of prompt, clear cut biological changes or health threats. As research has since proven, many of the fatal injuries in this field are never marked by “prompt, clear-cut biological changes” but instead appear only after a lengthy latency. While District officials made efforts to reduce the amount of exposure workers received from radiation and other toxic chemicals, there was not enough done to protect against the long term effects of increased exposure on an individual’s health.

At the Chicago Met Lab, scientists implemented the safeguards for plutonium work by putting linoleum on all the floors and having people use filter masks, rubber gloves, and outer protective clothing. Eating in the laboratories was not permitted, and Methods were developed to monitor the air in the labs for evidence of plutonium dust contamination. Similar safety procedures would be adopted at Los Alamos, Hanford, and Oak Ridge by the middle of 1944.

A procedure to detect the inhalation of plutonium dust using nose swabs was also introduced by 1944. The process involved a moist filter-paper swab which was inserted into the nostril and rotated, then the swab was spread out, dried, and read in an alpha detector. A reading of 100 counts per minute or higher was considered evidence of an exposure. Nose-swipe counts are still used for plutonium workers today and are part of the criteria used to decide when medical treatment for the worker, including prompt collection of urine samples, is necessary.

Safety monitoring equipment was also developed and implemented throughout the course of the Manhattan Project. They were used to monitor radiation and radioactive exposure, and prevent large-scale accidents at the reactors.

Radiation detectors were needed at Manhattan Project sites to delimit safe and dangerous areas and to monitor internal exposures to plutonium and other harmful radioisotopes. In the early stages of the Project, the Chicago Metallurgical Laboratory supplied the radiation detectors needed to monitor uranium and plutonium in the work environment. By 1944, however, a chronic shortage of radiation detectors led at least one site, Los Alamos, to begin a detector development program of its own.

The program was led by Richard Watts of the Electronics Group in the Physics Division. Watts and his team developed a number of alpha-particle detectors—culminating in the portable “Pee Wee”—named for its mere nineteen pounds, to detect radioactivity in the work environment.

The work done by Watts would set the stage for later work. After the war, the Los Alamos Laboratory emerged as a leader in radiation detection technology, developing some very special radiation detectors for monitoring internal exposures to radioisotopes. Wright Langham, the leader of the Radiobiology Group of the Health Division, organized a group to produce detectors that not only provided radiation protection but also had a significant impact in the fields of biology and nuclear medicine.

The Geiger-Müller counter was first conceived by German physicist Hans Geiger in 1908 and fully developed in 1928 after Walther Müller (a PhD student of Geiger) developed the sealed Geiger-Müller tube which could detect more types of ionizing radiation, making the device a more practical radiation sensor. A Geiger counter detects the emission of nuclear radiation — alpha particles, beta particles, or gamma rays — by the ionization produced in a low-pressure gas in a Geiger-Müller tube.

The Geiger counter was the most reliable radiation detector used during the Manhattan Project, and it has been a popular instrument for use in radiation dosimetry, health physics, experimental physics, the nuclear industry, geological exploration and other fields, due to its robust sensing element and relatively low cost.

Despite their reliability, Geiger counters still have their limitations. Because the output pulse from a Geiger-Muller tube is always the same magnitude regardless of the energy of the incident radiation, the tube cannot differentiate between radiation types. Geiger counters are also unable to measure extremely high radiation rates due to the “dead time” of the tube. This is an insensitive period after each ionization of the gas during which any further incident radiation will not result in a count, and the indicated rate is therefore lower than actual.

Scintillation counters also continued to develop during the course of the Manhattan Project. The technique was originally created in 1903 to detect alpha particles emitted by radium, and worked by counting the flashes of light produced when the alpha particles reacted with the zinc sulfide in the counter. But, the method fell into disuse by the 1930s because recording data was laborious and unreliable compared to the Geiger-Muller counters electronic output. The scintillator was revived with the development of the photomultiplier tube, an instrument that converts light into an electrical pulse. The innovation greatly improved the reliability of scintillation counters, allowing the number of counts could be recorded electronically. In addition, the discovery of a variety of new types of scintillators, liquid and solid, helped improve the rate of detection of alpha particles. Scintillation counting developed through the 1950s to produce the most versatile, sensitive, and reliable detectors of the time.

Liquid scintillation counters were another important innovation that were a marked improvement on existing techniques for the detection of tritium and other beta-emitting radioisotopes such as carbon-14 and phosphorus-32. Los Alamos

scientists became involved in these developments as research on the hydrogen bomb began during the early 1950s.

Reactors also needed to be monitored for the safety of the workers and surrounding communities. The B Reactor at Hanford had three different fail-safes to shut down the reactor to prevent a critical meltdown.

The first fail-safe, the SCRAM (Single Control Rod Axe Man) method, involved twenty-nine control rods with boron tips that hung from a rope above the reactor. If the reaction went out of control, the reactor operator would chop the rope and the rods would fall into the reactor and poison the reaction. The rope was soon replaced with an electric winch that could raise or lower the safety rods (but the acronym remained).

The second fail-safe method involved a boron solution acid that would be dumped into the reactor in the event that the safety control rods failed to descend into the reactor. The only problem with this method was that the solution, if used, would likely prevent the reactor from ever becoming operational again.

The boron solution acid was soon replaced with the Balls 3x system. This fail-safe involved tiny boron balls that could be dumped into the reactor to stop the reaction. This was a much more efficient method, because the balls could be retrieved and it did not permanently disable the reactor.

7.3.2. For the Secrecy of the Programme

The United States program to develop the atomic bomb, code-named the Manhattan Project, began in August 1942.

From the beginning, the need for security was paramount. The project had to be protected from sabotage and espionage and, equally important, the fact the U.S. was

working on such a program had to be kept under wraps at all cost. Early on, a Protective Security Section (PSS) handled personnel and information security, facility protection, and security education.

By February 1943, a more comprehensive counterintelligence program was warranted and Counter Intelligence Corps (CIC) agents Capt. Horace K. Calvert and Capt. Robert J. McLeod were assigned to the Manhattan Engineer District (MED) to organize the Intelligence Section.

More CIC personnel followed, with agents stationed at Oak Ridge, Tennessee; Chicago; St. Louis; Site Y (Los Alamos, New Mexico); and Berkeley, California. By August 1943, when the project transferred to the Corps of Engineers, the Intelligence Section merged with the PSS and established its headquarters at Oak Ridge. At this time, the Section assumed responsibility for every aspect of security within the MED. Four months later, on December 18, 1943, a special CIC Detachment, commanded by Lt. Col. William B. Parsons, was organized, and Lt. Col. John Lansdale became the chief of intelligence and security for the entire Manhattan Project.

In the early 1940s, Lansdale, a graduate of the Virginia Military Institute (VMI) and a U.S. Army Reserve officer, was a successful trial lawyer in Cleveland. He had turned down several calls for active duty before finally taking the advice of one of his VMI classmates to accept special duty within the War Department's Military Intelligence Division (MID).

Lansdale initially worked in the Investigation Branch, Counter Intelligence Group, reviewing investigative reports of prospective War Department employees. He eventually became chief of both the Investigation and Review branches of MID. Another one of his duties was to act as liaison between the PSS and the assistant chief of staff, Intelligence.

When the Manhattan Project transferred to the Corps of Engineers and the CIC Detachment activated, Lansdale had the background and connections to move effortlessly into the position as head of intelligence and security. Due to the criticality of his mission, Lansdale quickly became special assistant to Gen. Leslie Groves, the chief of the MED.

The CIC Detachment was initially comprised of 25 officers and 137 enlisted agents, each one hand-picked by Captains Calvert and McLeod. Over the next year, the Detachment grew to 148 officers and 161 enlisted agents. This included non-CIC military personnel with specific technical abilities critical to the security of the program. Detachment Headquarters was centralized at Oak Ridge, but personnel were placed on detached service in 11 branch offices around the nation. At times, these agents were so highly classified that they were referred to by code symbols and only the finance officer computing the pay of the agent knew his exact location.

Lansdale assumed full responsibility for all intelligence and security matters affecting the MED. In addition to preventing unintentional disclosure of information and infiltration by enemy agents, Lansdale's responsibilities included preventing fires and explosions, monitoring courier duties, protecting classified shipments, educating personnel about the importance of security measures, obtaining newspaper cooperation, and conducting 400,000 background investigations of potential personnel. His agents acted as bodyguards for the project's top scientists and went undercover to monitor local rumors about the various installations involved in the bomb development.

Lansdale also planned and executed the security measures for the 509th Composite Group, the special Army Air Forces' organization formed to deliver the bombs. Additionally, he was deeply involved in the Alsos Mission, an overseas task force that seized the technology and scientists involved in German atomic research.

The dropping of the atomic bomb on Hiroshima and Nagasaki, Japan brought about the end of World War II and saved the lives of thousands of U.S. and Allied troops who would have died in an invasion of Japan. The procedures put in place by Lansdale and his CIC Detachment led to the successful protection of the atomic bomb program, later called the "War's Best Kept Secret."

7.4. Methodology of Research Being Conducted

Invented in the early 1930s, cyclotrons ("atom smashers") accelerate atoms through a vacuum to induce collisions at very high speeds 25,000 miles per second via electromagnets. In 1936, John R. Dunning directed the development of Columbia University's own cyclotron in the basement of Pupin Hall. The project would take four years to complete. To produce the highest voltage possible, the team would employ a method developed by PhD student Herbert Anderson, using a pair of resonant concentric lines coupled to a high-power oscillator, marking the first use of this system.

In December 1938, German chemists Otto Hahn and Fritz Strassman detected barium by bombarding uranium with neutrons, subsequently making the discovery of nuclear fission. Hearing the news, Niels Bohr¹ travels to New York to find Columbia physicist Enrico Fermi. But when Bohr reaches campus, he doesn't find Fermi in his office, so he goes to the basement of Pupin Hall and encounters Anderson. Anderson recounted this experience years later in a 1965 interview with journalist Stephane Groueff:

Anderson: ...in January of 1939, Professor Bohr arrived with the news about fission. He arrived in New York and he came immediately to Columbia University, because he wanted to tell the news to Fermi. He did not find Fermi right away. He was looking for him and he found me instead. He was quite excited and he explained. He told everything to me.

Groueff: The first day he came to visit Columbia, you happened to be there?

Anderson: I happened to be there. He unburdened himself to me.

Groueff: You were just a young post graduate student?

Anderson: I was just starting to do my doctoral research.

That month, the Columbia experimental team consisting of John R. Dunning, Enrico Fermi, Herbert L. Anderson, Eugene T. Booth, G. Norris Glasoe, and Francis G. Slack conducted the first nuclear fission experiment in the US using Pupin Hall's very own cyclotron as they demonstrated the large energy release in the fission of uranium. A year later, Columbia physicists identified the exact fissionable material in uranium: the rare isotope uranium 235. To be used in a nuclear weapon, they reached the conclusion that uranium-235 must be separated from the more prevalent uranium-238 isotopes and concentrated. Dunning and Harold Urey then headed a Columbia team in the invention and perfection of the "gaseous diffusion" method of separating uranium isotopes.

This critical breakthrough served as the building block for the Manhattan Project's K-25 gaseous diffusion plant in Tennessee, which at the time of its completion, was the world's largest building. Decades after the Manhattan Project changed the course of history, the core component of the Columbia University cyclotron is now a part of the Smithsonian's National Museum of American History collection.

Check out a pdf copy of the National landmark filing for Pupin Hall and a pdf draft of the press release, written at the time of the landmark designation. Extensive documentation on the Pupin Physics Laboratories can also be accessed at the National Archives Catalog listing.

1 Willis Lamb, a distinguished faculty member from Columbia, claims he was the physicist who shared the news of the European fission experiment to Herbert Anderson, not Bohr.

8. Aftermath of the Project And the End of the War

The aftermath of World War II saw the rise of two global superpowers, the United States and the Soviet Union. The aftermath of World War II was also defined by the rising threat of nuclear warfare, the creation and implementation of the United Nations as an intergovernmental organization, and the decolonization of Asia, Oceania, South America, and Africa by European and East Asian powers, most notably by the United Kingdom, France, and Japan.

Once allies during World War II, the United States and the Soviet Union became competitors on the world stage and engaged in the Cold War, so called because it never resulted in overt, declared total war between the two powers. It was instead characterized by espionage, political subversion, and proxy wars. Western Europe was rebuilt through the American Marshall Plan, whereas Central and Eastern Europe fell under the Soviet sphere of influence and eventually behind an "Iron Curtain." Europe was divided into a U.S.-led Western Bloc and a Soviet-led Eastern Bloc. Internationally, alliances with the two blocs gradually shifted, with some nations trying to stay out of the Cold War through the Non-Aligned Movement. The Cold War also saw a nuclear arms race between the two superpowers, and part of the reason that the Cold War never became a "hot" war was that the Soviet Union and the United States had nuclear deterrents against each other, leading to a mutually assured destruction standoff.

As a consequence of the war, the Allies created the United Nations, an organization for international cooperation and diplomacy, similar to the League of Nations.

Members of the United Nations agreed to outlaw wars of aggression in an attempt to avoid a third world war. The devastated great powers of Western Europe formed the European Coal and Steel Community, which later evolved into the European Economic Community and ultimately into the current European Union. This effort primarily began as an attempt to avoid another war between West Germany and France through economic cooperation and integration, and a common market for important natural resources.

The end of the war opened the way for decolonization, as independence was granted to India and Pakistan (from the UK), Vietnam, Laos, Cambodia, French India and Vanuatu (from France), Indonesia (from the Netherlands), the Philippines (from the US), and several Arab nations from specific Mandates granted to European states by the defunct League of Nations. The State of Israel was also established following the disestablishment of the British-ruled Mandatory Palestine and the 1948 Palestine War. Nations in Sub-Saharan Africa achieved independence in the 1950s to 1970s.

The aftermath of World War II saw the rise of communist influence in East and Southeast Asia. The People's Republic of China was founded after the Chinese Communist Party (CCP) emerged victorious from the Chinese Civil War in 1949, and the First Indochina War was fought between the Viet Minh government and France after the Japanese retreat. The Korean War led to the division of the Korean Peninsula between the communist North Korea and the Western-aligned South Korea.

9. Procedure

What is a CC committee?

A crisis cabinet is a specialised committee that replicates decision-making during a crisis. Individual delegates or members inside the Crisis Cabinet provide directives to guide their actions and responses to the crisis. Delegates use directives to propose

specific courses of action, tactics, or policies to handle the situation. Depending on the committee structure and the course of actions currently happening in the committee, these directives can take many forms, such as Individual, Joint, Committee, Intelligence, Top Secret Directive, and Press Release.

Directive Types and How to Write Them

Individual Directive:

When an action is within your character's authority or is achievable due to their abilities, you write personal commands. To begin, there is a format for writing directives; so, you must state who is sending the directive and to whom it is being sent (from, to). Following that, whose cabinet is sending the directive, followed by the time and current date of the committee. Finally, the format of your directive and its title. That's all there is to it; this is the only format you need to know to write a directive. The only thing left is the content of your directive, and the way through which you compose it is fairly simple; You write it by answering the what, why, when, who, where, and, most crucially, how questions. You write the action you want to take by answering the WH questions and then elaborating and discussing it as much as possible to make your plan as thorough as feasible.

Also, whenever possible, employ the future tense. On paper, a directive looks like this:

From: Menelaus	Personal Directive	Date: 1192BC
May 3		

To: Related Authorities	Capturing Tiryns	Time: 12.34
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(Spartan Cabinet)

What: I will capture Tiryns with my 5,000 men stationed on the Tiryns frontline. My soldiers will kill and destroy any enemy forces they come across, as well as any

enemy military bases. Women and children in the city will not be murdered unless they attack the soldiers.

To reduce noise, the 5,000 agoge soldiers will be divided into 50 groups, with 200 soldiers per group. Each group will have a commander, and the commanders will be the best warriors among their groups. They will be well-armed with their hoplons, xiphos, and dorus (Spartan agoge soldiers' shields, spearheads, and small swords). Each group will apply the doctrine properly to face the fewest casualties. If needed, 3 soldiers from each group will bring supplies to their own group from the frontlines, and these 3 soldiers will be picked randomly from the commanders. They will use the safest route and avoid the enemy. Our men will take the safe paths suggested by our spies.

Why: Tiryns plays a crucial role in the war, and it must be captured in order to cut the enemy's supply lines.

When: Soldiers will charge at 02.00 a.m. to catch the enemy off guard.

Who: I will be operating this attack, and if I fall during the war, my right-hand man, Analus, will take over. 5,000 agoge soldiers will assault the enemy.

Where: 2,500 of my soldiers will charge from the southeast frontline, and the other 2,500 will charge from the west to capture Tiryns.

How*: Soldiers will check and control their weapons before charging. They will pray, remember how brave they are, and then honour their nation and gods by demolishing the enemy. They will not disobey their commander's orders and apply the doctrine as they say. To avoid being affected by attrition, our soldiers will study their geographical situation as well as the enemy's to use in their favour. Soldiers will use an offensive phalanx formation when I order them to charge, and they will slaughter each enemy troop they face. They will use the offensive formation until they face a larger enemy force to quickly capture as many critical areas as possible. If they face a larger enemy force, to be exact, 1.5 times larger than them, they will quickly change to a defensive phalanx formation and wait for recruitment whilst defending themselves. Their priority will be killing the enemy rather than cutting supply lines. In mountain areas, they will use the highlands in their favour and quickly oppress the enemy to finish them. Once they reach the city, their priority will be killing the cabinet members

of Tiryns. If possible, they will defenestrate them to entertain themselves. After the military bases and the city are captured, soldiers will go to the possible conflict areas to recruit other soldiers. Even though we sent spies before, our soldiers will be vigilant for any kind of trap. Their main objective is to capture the city, and for that purpose, they will sacrifice themselves without hesitation.

Joint Directive:

Directives written by more than one individual are considered joint directives. You write joint directives when you can only achieve your purpose in the directive by utilizing the authority of other cabinet members. Assume you are the commander of the army, and you want to take Warsaw. Yet you understand that conquering it without air superiority would be impossible, so you write a joint directive with the general of the air force. So it makes the "from:" part your name and the general of the air forces' name, and instead of a personal directive you write a Joint Directive. Everything else is the same.

Committee Directive:

A committee directive is written when you wish to use everyone's authority or when you are about to deliver your final directive (in most cases). Delegates frequently ask me, "How are we meant to write a committee directive with the other cabinet?" But that's just a common misunderstanding. You write the committee directive with your cabinet; the formal name is "Committee Directive," but it is basically a cabinet directive. So, simply writing a Committee Directive to the "from:" part will do.

Intelligence Directive:

You write intelligence directives when you want to acquire the necessary information about you or your country/cabinet. The format is exactly the same, except for the "WH Questions" part. For example:

From: Winston Churchill	Intelligence Directive	Date: 1942 May 3
To: Crisis Team	Our Troop Counts	Time: 16.21

(The Great Britain Cabinet)

How many troops does our country have? Do we possess any nuclear weapons? How many of our military factories are assigned to manufacture infantry weapons, and what kind of weapons are they producing?

Top Secret:

Top Secret directives are those that your chair is not allowed to read. You hand over your Top Secret directive to the admin. They are written precisely the same, but you must fold your paper and write "TOP SECRET" on the back side of it. The major reason for writing a Top Secret Directive is treason, a diabolical strategy to crash one's own cabinet or to switch sides. For instance, if a person secretly kills his cabinet members and becomes the dictator, the winning condition changes and only that person wins, whereas the cabinet loses. But I don't recommend writing Top Secret Directives unless you're planning on writing a brilliant 10-page long directive, because failing to do so will backfire much worse. You will earn your cabinet's distrust, and you may die and be given an insignificant character. Furthermore, updates to the Top Secret Directives are only sent to the person who sent them, unless they directly affect the other cabinet members.

Procedure of the Committee

The Crisis Cabinet (CC) procedure stands as a captivating and immersive experience that transcends the boundaries of traditional committee simulations. As delegates assume the roles of influential government officials and prominent leaders, they find

themselves thrust into the heart of rapidly evolving crises, where decisions carry weight and consequences are immediate. The SCC format showcases the art of crisis management, diplomacy, and collaboration, as participants grapple with fictional or historical scenarios that demand quick thinking and decisive action. Within the exclusive confines of a smaller, select committee, delegates navigate the intricacies of international conflicts, natural disasters, political upheavals, and military interventions, unveiling the true essence of leadership and strategic decision-making.

Tour De Table:

In Model United Nations (MUN) crises committees, "tour de table" is a term used to describe a specific procedure or phase of the committee session. It is a French term that translates to "turn of the table" in English.

During a tour de table, each delegate representing a country or entity in the committee takes turns speaking and sharing their country's or entity's stance, opinions, or proposed actions on the crisis at hand. This process allows all delegates to express their views and contribute to the discussion in a structured and organized manner.

Tour de table is often used at the beginning of a crisis committee session or at crucial points during the committee's work to gauge the positions and perspectives of each delegation. It helps set the stage for further debate, collaboration, and negotiation among delegates as they work to find solutions to the crisis. In a typical tour de table, the chair or moderator of the committee will go around the room or call upon delegates in a specific order to speak. Each delegate is given a limited time (usually around 1-2 minutes) to make their statement. This process ensures that all countries or entities have an equal opportunity to be heard and participate actively in the committee's discussions.

Semi-moderated Caucus: Unlike moderated caucuses, delegates in a semi-moderated caucus are allowed to speak without the chair's permission, as long as they do not interrupt other cabinet members and treat each other with respect.

Unmoderated Caucus:

In an unmoderated caucus, delegates are free to draft any kind of paper they want to achieve their goals, and support others. The majority of the cabinet's time should be spent on unmoderated caucuses rather than semi-moderated.

Unlike the General Assembly Committees, at a Joint Crisis Committee, the sessions proceed with Semi-Moderated caucuses and Unmoderated caucuses. The reason for that is to have a more smooth experience. Since the participants of a Joint Crisis Committee have already had a few experiences with how committees and MUN conferences proceed, semi and unmoderated caucuses are to advance the efficiency of debates, thus making crucial and quick decisions. It was mentioned earlier, but directives are the primary reason that a JCC's procedure is so unique. So later on with the sessions based on your chair's initiative you needn't take one or two semi-moderated caucuses before an unmoderated caucus to write directives, directly voting for an unmoderated caucus before moderates will be allowed to continue writing directives.

On a Joint Crisis Committee, after the roll call and Tour de Table, delegates take semi-moderated caucuses to discuss future actions, strategies and plans. After determining their move, delegates write down the designated act to put into practice. There isn't a General Speakers' List, resolution paper, or opening speech.

Imperial Edicts and Declarations (Ancient Press Releases)

In a contemporary committee, you would be using press releases to address the general public about any actions or situations. You can use press releases for propaganda to cleanse your own image, or black propaganda to tarnish somebody else's image. You can rally the commoner to join the army or try to break apart your opponent's military.

A press release is as its name implies, an act of publishing media. Press releases are important as a ruler is only as powerful as their flock is loyal.

Now in BCE the printing press or modern media obviously didn't exist. Yet rulers and people in positions of power have always utilized the power of controlling and speaking to the public. This can be done through Declarations and Edicts.

Imagine a marketplace in ancient times, you've seen it in a lot of movies. Some guy with a fanfare trots in on a horse and plays his tune, then proceeds to read loudly off a piece of parchment addressing the people in the name of the ruler. This is basically what a declaration is. The where, when and how is of course subject to your imagination and your creativity is the limit. You may choose to deliver speeches yourself at an important ceremony, or in front of your army to boost morale or approval of some sort. You may organize a whole event just for your declaration to be read. Just be creative.

An edict on the other hand is a little different. While a declaration is a mere giving of information, something you'd just listen to, an edict is an official order. A press release might say; Come and join your military now to help us beat our foul opponents and save your motherland from invasion!

An edict will say; By the ruler's authority, conscription has begun. Every man capable of holding a weapon must check-in for further procedure at the nearest military outpost in the next week. Citizens who fail to cooperate shall be punished.

Think of it as a "could/should" and a "must". Edicts are direct orders and might even be regarded as legislation. You might issue edicts to conscript members to your army, protect certain areas or properties, protect intellectual property, ban certain things or allow certain things, etc. They're basically laws in short.

10. Notes from the Committee Secretariat

In this committee, our main goal is to complete the Manhattan Project by building an atomic weapon. But this doesn't mean everything will go as planned. Throughout history we see betrayals, sabotages,, and most importantly, people who are bought by secret governments. While you are forcing your way to the gates of power, those who are against you will try to block your path.

You may need trials, police power, and spies, but don't forget, the one whom you try to find may be your most beloved person. Maybe your wife, your best friend, or the people you just met during the committee. So be careful who you trust. Money is power, and people are the pawns. Either you control them or you become one of them.

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